

THE SUMMER EE GLOW-UP

VOL. 4 | 10 RESOURCES THAT SHIP

Ten more resources and exactly how to use each one.

Volumes one to three taught you. This one gets a board in your hand.

```
buildlist $ start glow-up --vol 4 --resources 10  
loading board-level track ... ready.
```



READ THIS FIRST

This is volume four. Ten more of the best, with zero overlap from the first three lists.

Volume one got you building. Volume two went deep on theory and signals, and volume three handed you the channels and tools. This one is about the board itself: designing it, routing it, getting it manufactured, and bringing it up on the bench. Same format. Every resource has how to use it, the job it maps to, and a link. Tap any name to open it.

THE ONE RULE, STILL

Same rule as the first three guides. Do not try to do all 10. Pick 3 that point at the same job and go deep. Depth is what gets you hired and what makes content nobody can fake.

THE 10 RESOURCES

Each one with what it is, how to actually use it, and the job it maps to. Links are clickable.

01 KiCad

TOOL | CORE | free and open source

> [open kicad.org](https://www.kicad.org)

The full schematic-to-PCB toolchain, free forever, with no hobby tier and no seat limit.



HOW TO USE IT

- > Draw the schematic first and let the netlist drive the layout. Never lay out a board you have not drawn.
- > Make your first board two layers and boring. Finish it before you get clever.
- > Run DRC before every export. Those errors are the ones a fab charges you for.

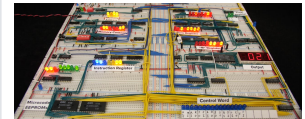
-> **The job this maps to:** PCB design engineer. The tool that turns a circuit on paper into a board you can hold.

02 Ben Eater

YOUTUBE | INTERMEDIATE | 8-bit computer on breadboards

> [open youtube.com/@beneater](https://www.youtube.com/@beneater)

He builds a working computer out of logic chips, one wire at a time, and explains every signal.



HOW TO USE IT

- > Follow the 8-bit series with parts in hand. Watching it is entertainment, building it is education.
- > Stop at each module, clock, registers, ALU, and make it work before you move on.
- > When something fails, find it on the scope before you watch his fix.

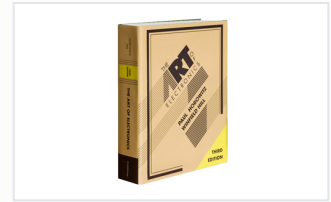
-> **The job this maps to:** Digital design and computer architecture. You stop treating a CPU as a black box.

03 The Art of Electronics

BOOK | ADVANCED | Horowitz and Hill, 3rd ed.

> [open artofelectronics.net](http://artofelectronics.net)

The reference working engineers keep on the bench, full of the judgement datasheets leave out.



HOW TO USE IT

- > Do not read it cover to cover. Look up the part you are using and read that section.
- > Read the bad-circuit examples. Knowing why a design fails beats collecting topologies.
- > Pair it with the Student Manual when you want problems that build real intuition.

-> **The job this maps to:** Generalist and analog engineer. The book that separates textbook answers from design.

04 Phil's Lab

YOUTUBE | ADVANCED | mixed-signal hardware design

> [open youtube.com/@PhilsLab](https://www.youtube.com/@PhilsLab)

Professional board design end to end: schematic, layout, manufacture, then firmware bring-up.



HOW TO USE IT

- > Watch one full board series start to finish before you route anything serious.
- > Copy his layout habits: ground planes, decoupling placement, where analog meets digital.
- > Use the STM32 and audio DSP videos to connect the board to the code running on it.

-> **The job this maps to:** Hardware design engineer. The gap between a student schematic and a manufacturable board.

05 EEVblog

YOUTUBE | CORE | Dave Jones, teardowns and bench

> [open youtube.com/@EEVblog](https://open.youtube.com/@EEVblog)

Thousands of teardowns, instrument reviews and blunt engineering opinion from a working designer.



HOW TO USE IT

- > Watch teardowns of gear from the field you want to work in. Real products teach real design.
- > Use the multimeter and scope reviews before you spend money on bench gear.
- > Treat the fundamentals playlist as the practical companion to your theory courses.

-> **The job this maps to:** Test and hardware engineer. Product-level judgement you cannot get from a lecture.

06 Nandland

FPGA | INTERMEDIATE | Verilog and VHDL from zero

> [open nandland.com](https://open.nandland.com)

A working FPGA engineer teaching the language, the tools, and which cheap board to start on.



HOW TO USE IT

- > Pick one language and stay in it all summer. Switching early costs you weeks.
- > Blink an LED in HDL on real hardware before you write a state machine.
- > Learn to read the timing report. In FPGA work that report is the truth.

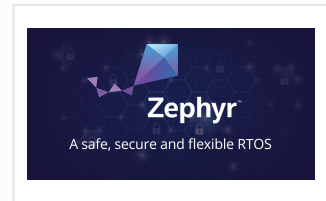
-> **The job this maps to:** FPGA and digital design engineer. A scarce skill, priced accordingly.

07 Zephyr Project

RTOS | ADVANCED | the Linux Foundation RTOS

> [open zephyrproject.org](https://open.zephyrproject.org)

The RTOS the embedded industry is standardising on, with real drivers, device tree and tooling.



HOW TO USE IT

- > Build and flash the blinky sample first. Getting the toolchain up is half the lesson.
- > Learn device tree properly. It is the part that confuses everyone and the part employers want.
- > Port one of your Arduino projects to Zephyr. The rewrite is where the concepts land.

-> **The job this maps to:** Embedded software engineer. The jump from sketches to production firmware.

08 Altium Education

TOOL | INTERMEDIATE | free for students

> [open education.altium.com](https://open.education.altium.com)

The industry-standard PCB tool, free for students, with a course that teaches board design properly.



HOW TO USE IT

- > Do the course project properly. It walks a real board from schematic to fabrication output.
- > Learn it after KiCad, not instead of it. Knowing both makes you portable on any team.
- > Keep the finished project files. They are portfolio pieces a recruiter can open.

-> **The job this maps to:** PCB design engineer. Fluency in the tool most hardware teams already run.

09 Microwaves101

REFERENCE | ADVANCED | RF and microwave encyclopedia

> [open microwaves101.com](https://microwaves101.com)

A working RF reference written by the engineers who design it, with the math and the rules of thumb.

MICROWAVES101.COM
The world's microwave information resource since 2001

HOW TO USE IT

- > Start with transmission lines and impedance matching. All of RF sits on those two.
- > Use the calculators while you design: microstrip width and matching networks in seconds.
- > Read the rules-of-thumb pages. That is the intuition RF interviews actually probe.

-> **The job this maps to:** RF and microwave engineer. Defense, radar and comms all hire from this skill set.

10 JLCPCB

SERVICE | CORE | boards made and shipped

> [open jlcpcb.com](https://jlcpcb.com)

Real fabricated boards for a few dollars, which is what turns your design from a file into hardware.



HOW TO USE IT

- > Order your first board before it is perfect. A flawed real board teaches more than a perfect file.
- > Read the capability specs before routing. Design to their minimum trace and drill from the start.
- > Get five made and give them away. Boards in other people's hands are proof you exist.

-> **The job this maps to:** Not a job, a multiplier. Shipping hardware is what makes you a hardware engineer.

THE 12-WEEK STACK

How to combine them so the summer actually compounds. Each phase ends with something you ship.

PHASE	DO THIS	YOU SHIP
FOUNDATION Weeks 1-4	KiCad open daily. The Art of Electronics as your reference. Ben Eater for the digital side.	A schematic you drew yourself and can defend line by line.
DEPTH Weeks 5-8	Phil's Lab for layout. Nandland or Zephyr for the digital track. Microwaves101 if you are going RF.	A routed board with a real BOM and a passing DRC.
BUILD Weeks 9-12	Order from JLCPCB, assemble it, and bring it up with EEVblog bench methods. Altium Education for the certificate.	An assembled board that powers up, and the log that proves you did it.

PICK YOUR LANE

Same move as last time. Three resources, one direction. EEVblog rides along no matter which you pick.

THE BOARD LANE

You want to design a PCB and hold the real thing.

- > KiCad
- > Altium Education
- > JLCPCB

Outcome: you design, order and assemble a board that works.

THE DIGITAL LANE

You want to understand machines from the gates up.

- > Ben Eater
- > Nandland
- > Zephyr Project

Outcome: you go from logic chips to HDL to production firmware.

THE ANALOG AND RF LANE

You want the signal side, and the layout that survives it.

- > The Art of Electronics
- > Phil's Lab
- > Microwaves101

Outcome: you design analog and RF that still works on a real board.

THE GLOW-UP IS NOT THIS PDF

Four guides in, you have no shortage of resources. The shortage is still reps. Pick your lane today and start phase one this week. The boards on your bench in September are the only proof that counts.

KEEP THE MOMENTUM

> THE BUILD LIST

The newsletter. Weekly build breakdowns and the projects you should not be able to make at home.

> FOLLOW THE BUILD

@1nska on Instagram. Daily build logs and the projects behind every guide.

Build in public. Ship by September.

Follow @1nska if you're building the future, not just studying for it.